SUMMER INTERNSHIP REPORT | CS3711

TRAIN RESERVATION SYSTEM — MUKAYA RAIL

UI/UX DESIGN OF A MOBILE TRAIN BOOKING EXPERIENCE

Submitted by

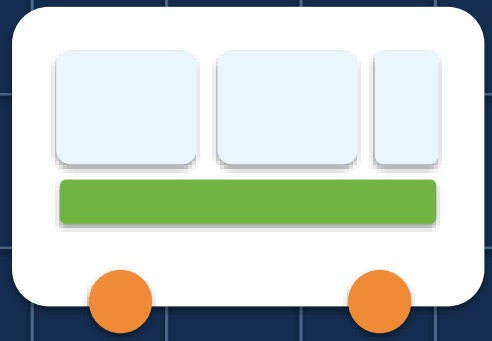
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MUKAYA RAIL

Book your journey easily

Introduction to Mukaya Rail

A passenger-first train reservation app concept

- Mukaya Rail is a mobile train reservation UI/UX case study.
- Designed during a summer internship at Media Wave Technologies, Tiruchirappalli.
- Covers the passenger journey from welcome and train search to booking confirmation.
- Includes an optional ticket-cancellation flow with refund details.
- Created using a user-centred design process: research, wireframes, visual design and prototyping.

PROJECT AT A GLANCE

RESEARCH
Study existing booking apps

DESIGN
Flows, wireframes, UI

PROTOTYPE
Linked screens in Figma

REVIEW
Navigation and usability

Problem Statement & Existing Experience

Common usability issues identified

- Some booking journeys require registration before users can explore.
- Long forms and strict formatting can interrupt the booking flow.
- Fare, fees and cancellation charges may appear late in the journey.
- Users can be uncertain about payment status or refund amount.

Design response in Mukaya Rail

PAIN POINT	DESIGN APPROACH
Account first	Guest or sign-in choice
Unclear journey	Step-by-step progress
Late cost visibility	Order summary before payment
Payment uncertainty	Clear payment-success screen
Cancellation doubt	Refund details before confirmation

The project addresses these issues through interface design; it does not implement live railway services.

Project Objectives & Scope

OBJECTIVES

- Study existing train booking interfaces
- Plan user flow and information architecture
- Create wireframes, reusable UI components and a prototype

IN SCOPE

- Welcome and booking mode
- Login / sign-up and train search
- Train selection and passenger details
- Order summary, payment success and cancellation

DESIGN DELIVERABLES

- Eleven connected mobile screens
- Consistent colours, typography, cards and buttons
- Interactive Figma prototype
- Review and refinement based on feedback

OUT OF SCOPE

- Backend services and live seat availability
- Real payment gateway integration
- Railway operator system integration
- Production database implementation

Focus: a clear, consistent passenger-facing experience—not a deployed booking backend.

Proposed Design & Key Features

The prototype links the main screens into one guided booking journey.

Flexible entry

Guest booking or sign in

Train discovery

Search and select trains

Passenger details

Input fields and correction messages

Cost clarity

Order summary before payment

Booking feedback

Payment-success screen with PNR

Cancellation clarity

Refund details before confirmation

11 DESIGNED SCREENS • INTERACTIVE PROTOTYPE

Technologies & Design Tools

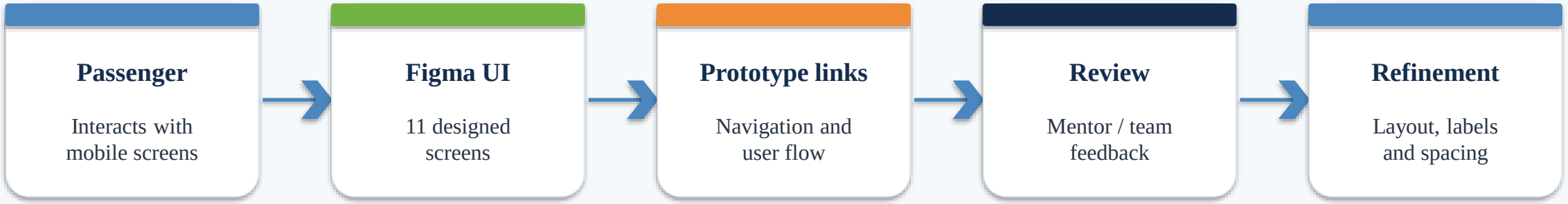
Tools explicitly listed in the internship report

Figma	Primary tool for UI design, wireframing, reusable components, auto layout and interactive prototyping.
Google Fonts / system fonts	Typography choices to support readability.
Icon sets	Consistent iconography across the interface.
PNG / PDF export	Sharing final screens for review and documentation.

No programming language, database, or backend framework is specified as part of this UI/UX project.

System Architecture & Data Design

Design-level architecture represented by the Figma prototype



DATABASE / BACKEND STATUS

A database schema, live seat-availability service, payment gateway and railway-system integration are not implemented or specified in the report. The screens are a UI/UX prototype.

This diagram describes the documented design workflow, not a production software architecture.

Implementation & System Workflow

Four-week internship workflow

WEEK 1

Understand

Company orientation, study booking apps, list user needs and screens.

WEEK 2

Plan

Map the user flow and create low-fidelity wireframes.

WEEK 3

Design

Create the design system and design welcome, account, search and train screens.

WEEK 4

Prototype & review

Design passenger, summary, payment and cancellation screens; link and refine prototype.

User flow: Welcome → Guest / Sign in → Search Train → Select Train → Passenger Details → Order Summary → Confirm Payment → Payment Success / Cancel Ticket

Results, Conclusion & Future Scope

RESULTS / REVIEW

- Prototype navigation reviewed
- Search and train-selection flow checked
- Passenger details and validation messages reviewed
- Order summary and payment flow checked
- Cancellation flow reviewed

Report records checks as verified.

CONCLUSION

- Eleven screens form a consistent booking journey
- Guest booking reduces entry friction
- Fare and refund details are shown before confirmation
- Feedback helped refine alignment, labels and spacing

FUTURE SCOPE

- Develop a production application
- Connect backend and live seat availability
- Integrate a real payment gateway
- Integrate railway operator services

These are future possibilities, not current implementations.

Outcome: a user-centred, interactive UI/UX prototype for train reservation.

References & Thank You

REFERENCES

- Nivetha S., UI/UX Design of Train Reservation System – Mukaya Rail Mobile Application, Summer Internship Report, 2026–2027.
- Figma — design, wireframing and prototyping tool referenced in the report.
- Google Fonts / system fonts and icon sets, as listed in the report.

THANK YOU

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Questions & Discussion

Mukaya Rail
“Book your journey easily”

CS3711 | Academic Year 2026–2027